

OBC v1.2 Datasheet

High-Level Overview

- Features an STM32G431RBT6 microcontroller^[1]
 - 32-bit Arm Cortex-M4 processor with floating-point support
 - Maximum operating frequency of 170 MHz
 - 128 kB of internal Flash memory
 - 32 kB of SRAM
 - LQFP-64 package
- Uses a 24 MHz High-Speed External (HSE) crystal and a 32.768 kHz Low-Speed External (LSE) crystal.
- Supports communication and control through SPI, I²C, UART, CAN, GPIO, timer, and analog connections.

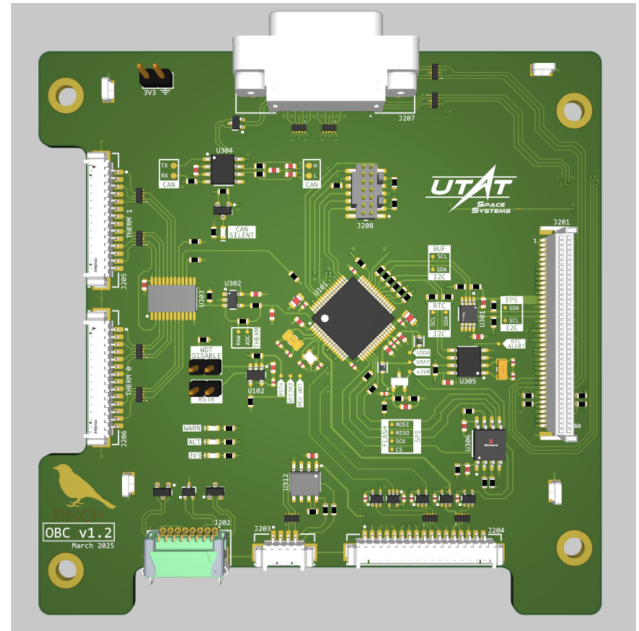


Figure 1: OBC v1.2

Introduction

The OBC v1.2 serves as the primary processing and coordination board for the spacecraft. It runs onboard firmware, communicates with other spacecraft subsystems, monitors system signals, and controls board-level functions.

This datasheet documents the main electrical interfaces, functional blocks, connectors, and implementation details of the board. It is intended to preserve the design for future team members and support board review, manufacturing, bring-up, and future redesign work.

Device Overview

The block diagram below represents the high-level connection between the MCU and the spacecraft bus.

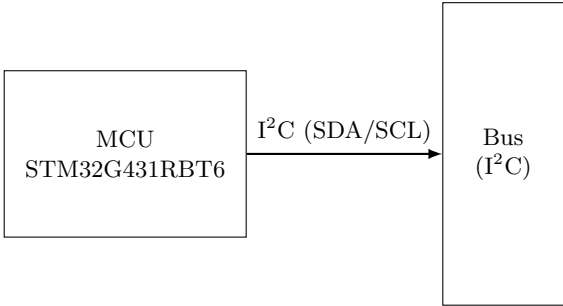


Figure 2: High-level block diagram of the MCU and spacecraft bus

Specifications

TODO: Cover electrical specifications such as voltage ratings, operating conditions, and board-level power requirements.

Interconnects and Pinout

TODO: Add an introduction to the OBC interconnects and pinout.

Bus

The table below maps the J201 connector pins found in the Protel netlist to their corresponding net labels.

Pin	Net
<i>Netlist table unavailable: netlisthelper.tex was not found.</i>	

Power

TODO: Describe the power connections available through the bus connector.

EPS Connection

TODO: Describe the bus connections between the OBC and EPS.

GPIO

TODO: Describe the general-purpose signals routed through the bus.

GNSS

TODO: Describe the GNSS signals routed through the bus.

CAN Bus

TODO: Describe the CAN bus connections and corresponding pins.

RS-422

TODO: Describe the RS-422 connections and corresponding pins.

RS-485

TODO: Describe the RS-485 connections and corresponding pins.

PAY Debug

TODO: Describe the payload debug connections and corresponding pins.

ADCS

TODO: Specify the ADCS connector pinout and reference its layout position. Add an image of the connector and the corresponding BOM part. Cover the protection circuitry in the corresponding subsection.

JTAG

TODO: Specify the JTAG pinout and reference its layout position. Add an image of the connector and the corresponding BOM part.

SRS-4

TODO: Specify the pinout and reference its layout position. Add an image of the connector and its corresponding BOM part.

Thermistors

TODO: Specify the thermistor connector pinout and reference its layout position. Add an image of the connector and its corresponding BOM part.

TODO: Describe the thermistor multiplexers used by the OBC, their use case, part numbers, and the associated operational-amplifier circuitry.

Umbilical

TODO: Specify the umbilical connector pinout and reference its layout position.

Functional Description

MCU

The OBC uses the STM32G431RBT6 from STMicroelectronics as its main processing device. It contains a 32-bit Arm Cortex-M4 processor with floating-point support, 128 kB of internal Flash memory, and 32 kB of SRAM in an LQFP-64 package^[1].

The MCU runs the onboard firmware, monitors system signals, controls board-level functions, and communicates with the other spacecraft subsystems. The schematic assigns SPI, I²C, UART, CAN, ADC, timer, and GPIO connections to the MCU^[2]. Detailed operation of the connected peripherals is covered in their respective sections rather than repeated here.

Internal meeting notes indicate that compatibility with the Zephyr Real-Time Operating System was an important consideration during MCU selection. The notes also show that the team preferred an STM32 device and wanted the simplest MCU that could still meet the firmware and interface requirements^[3]. The complete MCU trade study was not located, so the final selection reasoning should still be confirmed with the original design lead.

Power and Analog Reference

The MCU is powered from the board's regulated 3.3 V rail. Its digital supply pins are locally decoupled using four 0.1 μ F capacitors, together with 1 μ F and 4.7 μ F capacitors^[2]. The smaller capacitors provide local high-frequency decoupling, while the larger capacitors provide additional nearby charge storage.

The analog supply, VDDA, is connected to the 3.3 V rail through a ferrite bead and has separate local decoupling capacitors. This helps limit the amount of digital switching noise reaching the MCU's analog circuitry^[2].

VREF+ is supplied by a dedicated 3.0 V reference circuit based on an ATL431 programmable shunt reference. The reference is separately filtered before reaching the MCU, providing a defined voltage reference for analog-to-digital conversion^[2].

The VBAT pin is connected to the same regulated 3.3 V rail in this board revision rather than to a separate backup battery supply^[2].

Reset and Fault Recovery

The NRST pin is normally held high using a 10 k Ω pull-up resistor. A 10 nF capacitor between NRST and ground forms an RC network that helps filter brief disturbances on the reset line^[2]. A manual reset connection is also included for development and board testing.

The reset network can additionally be controlled by the debug interface and an external STWD100 watchdog supervisor. If the MCU stops providing the expected watchdog signal, the external watchdog can assert the reset line and restart the MCU^[2].

The STM32G431 also contains independent and window watchdog peripherals^[1]. The external watchdog therefore provides an additional recovery path that is independent of the MCU's internal firmware execution.

Programming and Debugging

Firmware programming and hardware-level debugging are provided through the Serial Wire Debug interface. SWDIO and SWCLK are routed from the MCU to the OBC debug interface, allowing an external debugger to program internal Flash, inspect memory and registers, pause execution, and step through firmware^[1,2].

The design also exposes SWO for optional trace output and includes a reset connection that allows the debugger to issue a hardware reset. A separate UART connection is routed to the Ground Support Equipment interface for firmware-controlled communication during board bring-up, testing, and integration^[2].

External Interfaces

The MCU communicates with the rest of the OBC through SPI, I²C, UART, CAN, GPIO, timer, and analog connections. These include interfaces to the EPS, ADCS, external Flash memory, thermal-monitoring circuitry, GNSS signals, spacecraft bus, and debug circuitry^[2].

Only the MCU-side purpose and allocation of these interfaces are covered here. Their detailed implementation is documented in the corresponding subsystem sections.

External Clocks

The MCU is provided with both high-speed and low-speed external clock sources.

A 24 MHz crystal is connected between OSC_IN and OSC_OUT, with two 5.1 pF load capacitors^[2]. This provides the High-Speed External clock source. The STM32G431 supports HSE crystals between 4 MHz and 48 MHz, so the selected 24 MHz crystal is within the supported range^[1].

A separate 32.768 kHz crystal is connected between OSC32_IN and OSC32_OUT, with two 3.9 pF load capacitors^[2]. This provides the Low-Speed External clock source and can be used for accurate low-frequency timing, including operation of the integrated real-time clock^[1].

The crystal, oscillator pins, and load capacitors form an oscillator feedback circuit. The crystal determines the frequency, while the amplifier inside the MCU sustains the oscillation through OSC_IN and OSC_OUT. STMicroelectronics recommends placing the crystal and load capacitors close to the oscillator pins to reduce distortion and improve startup behaviour^[1].

External Flash Memory

The OBC includes an SST26VF064B external serial Flash memory device. It provides 64 Mbit, or 8 MB, of additional non-volatile storage beyond the MCU's internal Flash memory^[4].

The Flash memory is connected to the MCU through SPI1 using chip select, serial clock, MOSI, and MISO signals^[2]. SPI provides a relatively simple interface for transferring stored data without requiring a parallel address or data bus.

The intended contents of the external Flash were not identified in the available design notes. This section therefore does not assume whether the memory was intended for telemetry, firmware images, logs, configuration data, or another application.

RTC

Timekeeping is provided by the real-time clock peripheral integrated within the STM32G431RBT6 rather than by a separate external RTC device^[1].

The external 32.768 kHz crystal connected to OSC32_IN and OSC32_OUT can provide the low-frequency clock used by the RTC^[1,2]. The use of this crystal provides a more accurate low-frequency reference than relying only on the MCU's internal low-speed RC oscillator.

The MCU's VBAT pin supplies the RTC and backup domain when a separate backup source is available^[1]. In this board revision, VBAT is tied to the same regulated 3.3 V rail as the rest of the MCU^[2]. The design therefore does not include an independent battery-backed supply that would maintain the RTC after the board's 3.3 V rail is removed.

The exact use of the RTC within the OBC firmware was not confirmed in the available documentation.

Mechanical and Layout

TODO: Add an isometric view of the board showing the maximum component height, board dimensions, and drill-hole locations.

TODO: Add a description of the polygon pours used for power distribution and thermal management.

Manufacturing

Bill of Materials

TODO: Add or reference the board Bill of Materials.

Test Points

TODO: Describe the available test points within the design and cross-reference the corresponding functional subsections.

Bring-Up Log

TODO: Specify the tests that have been conducted and reference the corresponding Notion documentation.

References

- [1] STMicroelectronics, “STM32G431x6, STM32G431x8, STM32G431xB: Arm Cortex-M4 32-bit MCU+FPU, 170 MHz/213 DMIPS, up to 128 KB Flash, 32 KB SRAM, rich analog, math accelerator,” Datasheet DS12589, Rev. 6, Oct. 2021.
- [2] University of Toronto Aerospace Team, Space Systems Division, “OBC v1.2 schematic: STM32G431 MCU and associated hierarchical sheets,” internal Altium design files, unpublished, 2026.
- [3] University of Toronto Aerospace Team, Space Systems Division, “FDIR Sync with EPS and Firmware,” internal meeting notes, unpublished, date not identified.
- [4] Microchip Technology Inc., “SST26VF064B/SST26VF064BA 2.5 V/3.0 V 64-Mbit Serial Quad I/O Flash Memory,” Datasheet DS20005119, Rev. G.